```sql
#Q1. The marketing team wants to understand the size of our customer base in each country.
#Can you provide the number of customers in each country and identify the countries with the largest customer base?
select country,count(customer_id) as no_customers from customers group by country order by no_customers desc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

country	no_customers
Italy	69
Spain	63
Morocco	59
France	56
Germany	53

--

```sql
#Q2. The operations manager wants an overview of our order pipeline.
#Can you show how many orders are currently in each status and identify the most common order status?
select status, count(*) as order_count from orders group by status order by order_count desc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

country	no_customers
Italy	69
Spain	63
Morocco	59
France	56
Germany	53

```sql
#Q3. The operations team is currently investigating cancelled orders.
#Can you retrieve the cancelled orders and show their order IDs and order dates?
select order_id,order_date from orders where status='cancelled';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

order_id	order_date
8	2024-06-23
34	2024-11-23
42	2024-04-01
45	2024-04-05
76	2024-10-19
82	2024-12-07
100	2024-10-07
104	2024-10-04
106	2024-01-04
107	2024-06-02
108	2024-01-29
118	2024-02-11
120	2024-02-10

orders 5 ×

```sql
#Q4. The finance team wants to review high-priced transactions.
#Can you identify the order items where the unit price is above ₹100 and display the highest-priced items first?
select * from order_items where price>100 order by price desc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

order_id	product_id	quantity	price
22	25	4	119
892	19	3	119
28	48	4	119
361	20	4	119
523	47	1	119
41	16	1	119
317	4	1	119
864	2	5	119
326	8	5	119
795	15	3	119
76	10	3	119
925	14	3	119
73	13	1	119
905	44	4	119

```sql
#Q5. The marketing team wants to identify the dates on which we acquired the most customers.
#Can you show the number of customer signups for each signup date and identify the busiest signup dates?
select signup_date, count(*) as count from customers group by signup_date order by count desc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

signup_date	count
2023-08-09	4
2023-06-12	4
2023-06-07	4
2024-04-07	3
2024-01-03	3
2023-09-01	3
2024-01-25	3
2024-02-15	3
2023-12-24	3
2023-05-07	3
2023-12-15	3
2023-01-29	3
2023-06-03	3
2023-05-15	2

```sql
#Q6. The operations manager wants to investigate orders placed on a particular date.
#Can you retrieve all orders placed between august to october and arrange them by order ID?
select * from orders where order_date between '2024-08-01' and '2024-10-31' order by order_id;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

order_id	customer_id	order_date	status
1	103	2024-10-31	Completed
2	271	2024-10-12	Completed
6	21	2024-09-13	Completed
12	152	2024-08-09	Completed
16	294	2024-09-20	Completed
20	22	2024-10-14	Completed
21	253	2024-09-22	Completed
25	170	2024-10-14	Completed
31	55	2024-09-03	Completed
33	131	2024-09-11	Completed
36	167	2024-09-05	Completed
37	274	2024-08-26	Completed
38	89	2024-08-16	Completed
43	92	2024-09-20	Completed

```sql
#Q7. The product manager wants a quick overview of our pricing.
#Can you report the minimum, maximum and average recorded unit price across our transactions?
select min(price) as min, max(price) as max, avg(price) as avg from order_items;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

min	max	avg
10	119	64.8270

```sql
#Q8. Management wants to know whether our orders are concentrated in a particular status.
#Can you identify the percentage of orders represented by each status?
select status, round(count(*) * 100/ (select count(*) from orders),1 )as percentage from orders group by status;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

status	percentage
Completed	80.5
Cancelled	10.3
Returned	9.2

```sql
#Q9. The finance manager wants to know the total sales value generated from the available order items.
#Can you calculate the overall sales value using quantity and unit price?
select sum(quantity*price) as tot_sales from order_items;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

tot_sales
384853

```sql
#10. The sales manager wants to compare the financial performance of our product categories.
#Can you calculate total sales value for each category and display the categories from highest to lowest?
select p.category,sum(oi.quantity*oi.price) as tot_sales from products as p
join order_items as oi on p.product_id=oi.product_id group by p.category order by tot_sales desc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

category	tot_sales
Hair	126301
Makeup	103662
Body	88177
Skin	66713

```sql
#11. We're planning a customer loyalty program.
#Can you identify the 10 customers who have generated the highest sales value?
select c.customer_id, sum(oi.quantity*oi.price) as sales from customers as c
join orders as o on c.customer_id=o.customer_id
join order_items as oi on o.order_id=oi.order_id
group by c.customer_id order by sales desc limit 10;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

customer_id	sales
152	5388
13	5048
39	4564
231	3769
25	3509
144	3420
172	3416
76	3330
192	3275
268	3264

```sql
#12. The product team wants to identify the products with the strongest demand.
#Can you find the top 10 products based on total quantity sold?
select p.product_id,p.product_name, sum(oi.quantity) as tot_qty from products as p
join order_items as oi on p.product_id=oi.product_id
group by p.product_id,p.product_name order by tot_qty desc limit 10;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

product_id	product_name	tot_qty
45	Product_45	171
47	Product_47	163
6	Product_6	157
25	Product_25	156
23	Product_23	156
42	Product_42	154
32	Product_32	152
16	Product_16	151
4	Product_4	144
26	Product_26	144

```sql
#13. The sales manager wants to know which individual products contribute the most to revenue.
#Can you display products based on their total sales value like lowest to highest?
select p.product_id,p.product_name, sum(oi.quantity*oi.price) as tot_sales from products as p
join order_items as oi on p.product_id=oi.product_id
group by p.product_id,p.product_name order by tot_sales;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

product_id	product_name	tot_sales
46	Product_46	5563
24	Product_24	5683
33	Product_33	5856
37	Product_37	5960
10	Product_10	6017
9	Product_9	6060
2	Product_2	6086
5	Product_5	6180
27	Product_27	6313
49	Product_49	6349
29	Product_29	6416

```sql
#14. The finance team wants to understand the typical value of an order.
#Can you calculate the average sales value per order?
select avg(sum_sales) as avg_sales from (select o.order_id, sum(oi.quantity*oi.price) as sum_sales from orders as o
join order_items as oi on o.order_id=oi.order_id group by o.order_id) as x;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

avg_sales
446.9837

```sql
#15. The CRM team wants to identify customers who have placed multiple orders.
#Can you find customers with more than one order so that we can analyze repeat purchasing behavior?
select o.customer_id, count(*) as order_count from orders as o
join customers as c on c.customer_id=o.customer_id
group by o.customer_id having order_count>1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customer_id	order_count
103	4
271	4
107	2
72	2
189	3
21	2
122	6
215	6
100	4
152	7
131	4
150	4

```sql
#16. The marketing manager wants to create a VIP customer segment.
#Can you identify customers whose total spending exceeds a specified business threshold (i.e, 2500)
select c.customer_id, sum(oi.quantity*oi.price) as tot_spend from customers as c
join orders as o on c.customer_id=o.customer_id
join order_items as oi on o.order_id=oi.order_id
group by c.customer_id having tot_spend > 2500;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customer_id	tot_spend
152	5388
113	3084
289	3012
220	3253
13	5048
268	3264
144	3420
51	2738
39	4564
76	3330
25	3509

```sql
#17. The product team wants to compare pricing across our product categories.
#Can you identify the minimum and maximum recorded unit price for each category, along with the number of units sold in that category?
select p.category, min(oi.price) as min_unit_price,
max(oi.price) as max_unit_price, sum(oi.quantity) as units_count from products as p
join order_items as oi on oi.product_id=p.product_id group by p.category;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

category	min_unit_price	max_unit_price	units_count
Hair	10	119	1971
Makeup	10	119	1623
Body	10	119	1363
Skin	10	119	1018

```sql
#Can you classify each customer into the appropriate spending segment?
with details as(
select c.customer_id, sum(oi.quantity*oi.price) as tot_spend from customers as c
join orders as o on c.customer_id=o.customer_id
join order_items as oi on o.order_id=oi.order_id
group by c.customer_id
)
select customer_id , case
when tot_spend <1000 then 'Low'
when tot_spend between 1000 and 3000 then 'Medium'
else 'High'
end as spending_segment from details;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customer_id	spending_segment
170	Medium
256	Low
54	Medium
152	High
159	Low

```sql
#Q19. Each category manager wants to know which products are performing best within their category.
#Can you identify the top three products in every category based on total sales value?
with cat_detail as(
select p.category, p.product_name, sum(oi.quantity*oi.price) as tot_sales,
row_number() over( partition by p.category order by sum(oi.quantity*oi.price) desc) as rs
from products as p join order_items as oi on p.product_id=oi.product_id
group by p.category,p.product_name)
select * from cat_detail where rs<=3;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

category	product_name	tot_sales	rs
Body	Product_23	10821	1
Body	Product_45	10316	2
Body	Product_6	9547	3
Hair	Product_25	10649	1
Hair	Product_47	10539	2

```sql
#Q20. Management wants to identify customers who are more valuable than the average customer.
#Can you find customers whose total spending is higher than the average spending across all customers?
with spending as(
select c.customer_id,c.country, sum(oi.quantity*oi.price) as tot
 from customers as c join orders as o on c.customer_id=o.customer_id
 join order_items as oi on oi.order_id=o.order_id group by c.customer_id, c.country)
 select*from spending where tot> (select avg(tot) from spending);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customer_id	country	tot
170	Morocco	1978
54	Morocco	1847
152	Morocco	5388
60	Spain	1930
95	Spain	1858

```sql
#21. The marketing team wants to prioritize customers according to their contribution to sales.
#Can you rank customers from highest to lowest based on their total spending?
select c.customer_id,c.country, sum(oi.quantity*oi.price) as tot,
dense_rank() over(order by sum(oi.quantity*oi.price) desc) as rs
from customers as c join orders as o on c.customer_id=o.customer_id
join order_items as oi on oi.order_id=o.order_id
group by c.customer_id, c.country;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customer_id	country	tot	rs
152	Morocco	5388	1
13	Italy	5048	2
39	Morocco	4564	3
231	Morocco	3769	4
25	Italy	3509	5
144	France	3420	6
172	Italy	3416	7
76	Spain	3330	8
192	Italy	3275	9
268	France	3264	10
220	France	3253	11

```sql
#22. The category managers want to identify their strongest individual product.
#Can you find the single highest-revenue product within every product category?
with strong_pdct as (
select p.product_id,p.product_name,p.category ,row_number() over( partition by category order by sum(oi.quantity*oi.price) desc) as r
from products as p join order_items as oi on p.product_id=oi.product_id group by p.product_id,p.product_name,p.category)
select * from strong_pdct where r=1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

product_id	product_name	category	r
23	Product_23	Body	1
25	Product_25	Hair	1
16	Product_16	Makeup	1
13	Product_13	Skin	1

```sql
#23. Management wants to understand how dependent the business is on its highest-value customers.
#What percentage of our total sales value is generated by the top 10 customers?
select sum(tot)*100/(select sum(quantity*price) from order_items) as percentage from
(select c.customer_id, sum(oi.quantity*oi.price) as tot
from customers as c join orders as o on o.customer_id=c.customer_id
join order_items as oi on oi.order_id=o.order_id group by c.customer_id
order by tot desc limit 10) as y;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

percentage
10.1293

```sql
#24. Management wants a reusable customer sales summary for future reporting.
#Can you create a SQL view containing each customer's total number of orders, total quantity purchased and total spending?
create view sales_summary as select c.customer_id,count(distinct o.order_id) as order_count,
sum(oi.quantity) as tot_qty,sum(oi.quantity*oi.price) as tot_spend from customers as c join orders as o on o.customer_id=c.customer_id
join order_items as oi on oi.order_id=o.order_id group by c.customer_id;
select * from sales_summary;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customer_id	order_count	tot_qty	tot_spend
1	8	51	2632
2	3	28	2075
3	1	2	180
4	4	12	774
5	7	47	3014
6	3	25	1597
7	2	16	1087
8	2	12	581
9	3	18	961

```sql
#25. The customer-support team frequently needs to retrieve a customer's purchase history.
#Can you create a stored procedure that accepts a customer ID and returns all orders and products purchased by that customer?
delimiter //
create procedure purchase_history(in id int)
begin
   select c.*,o.order_id,order_date,o.status,p.*,oi.quantity,oi.price from customers as c join orders as o on o.customer_id=c.customer_id
join order_items as oi on oi.order_id=o.order_id
join products as p on p.product_id=oi.product_id where c.customer_id=id;
end //
delimiter ;
call purchase_history(45);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customer_id	country	signup_date	order_id	order_date	status	product_id	product_name	category	quantity	price
45	Italy	2023-10-11	132	2024-10-13	Completed	36	Product_36	Skin	3	82